

10 Boron

Boron supports growing points, cell walls, pollen function, and fruit set. The range between too little and too much is narrow, and irrigation water can be a major source.

Why it matters

- **Growing tissues.** Boron supports cell-wall formation, new root and shoot growth, pollen germination, and reproductive development.
- **Narrow safety margin.** Deficiency and toxicity can both damage growth. Small unmeasured applications can create uneven hot spots.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists roughly 10 to 20 ppm in the deficient category, 20 to 100 ppm as adequate, and 100 to 250 ppm as excess.
Use with care	The guide and fertilization chapter should be read together. Low or borderline B deserves investigation, not automatic application.

Main soil and environmental controls

Factor	What it can mean in the root zone
Irrigation water	Water B accumulates over time and should be measured directly. Sodium borate sources may also add sodium.
Texture and organic matter	Sandy soils may leach B; clay, oxides, and organic matter can hold it. Release changes with wetting, drying, pH, and decomposition.
pH	Availability often declines as pH rises, but soil tests are method- and site-dependent and cannot be interpreted by pH alone.
Drought salinity and leaching	Drying concentrates salts and increases exposure. Leaching can reduce soluble B only when adequate low-B water and drainage are available.
Root zone	Shallow wetting, poor uniformity, waterlogging, and root disease create uneven uptake and make tissue patterns harder to read.

Field clues and checks

- **Deficiency clues.** Distorted young leaves, shot holes, loss of growing points, poor flowering, and misshapen fruit can occur, but other causes must be excluded.
- **Toxicity clues.** Older-leaf yellowing, tip or margin burn, and defoliation can overlap chloride, sodium, drought, and general salinity injury.
- **Test all sources first.** Use irrigation-water, soil, and standardized leaf tests plus the history of fertilizers, composts, sprays, and amendments.

Do not use a bloom claim, one leaf number, or a model interaction as the sole basis for boron treatment.

DST interpretation Phase 1 P by B and Zn by B signals remain research candidates. Because boron has a narrow safety margin, no model signal should trigger application without confirming water, soil, leaf, and field evidence.